

Impact Force Location and Intensity Identification Using Joint-Position Sensors in Humanoids

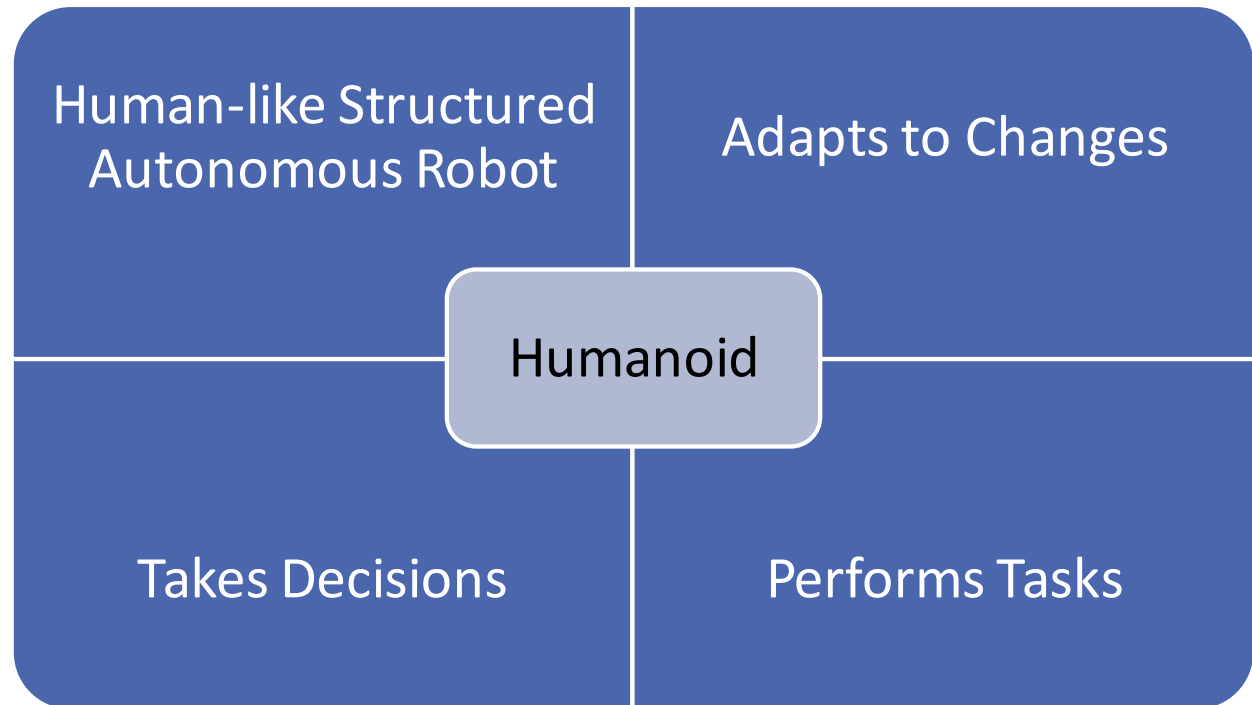


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Outline

- ◀ Introduction
- ◀ Motivation , Hypothesis and Proposed Solution
- ◀ Literature Review and Sate of the Art
- ◀ Methodology Of The Work
- ◀ Machine Learning Algorithm
- ◀ Discussion of Results
- ◀ Conclusion
- ◀ Relevance of the Work

Introduction





Motivation

Humanoids are designed
to imitate human
behaviors



Hypothesis

Giving the humanoid the
ability to evaluate and
allocate the force
enhances the stabilization



Solution

Evaluation and allocation
of an impact force based
on the vibrations in all
joints using Machine
Learning

Literature Review

ATLAS

The world's Most Dynamic Humanoid Robot

By: Boston Dynamics

July 11, 2013

1.5 m High

Weighs 80 Kg

28 Hydraulic Joints

Moves with Speed of 1.5 m/s

Controlled by an On-Board Computer

Perceives the Environment via LIDAR

Boston Dynamics



State of the Art

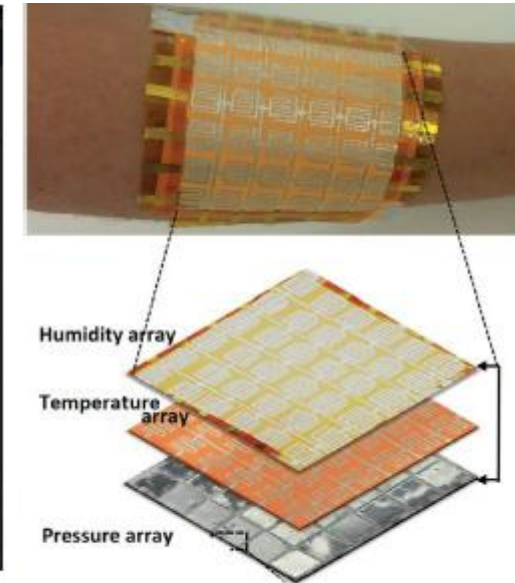
MULTITASKING HUMANOID FOR SURVEILLANCE & MONITORING

BY: ASHWIN DHANDE (2018)

Artificial skin used to detect forces along with other parameters in harsh environments



ROBONAUT DESIGNED BY NASA



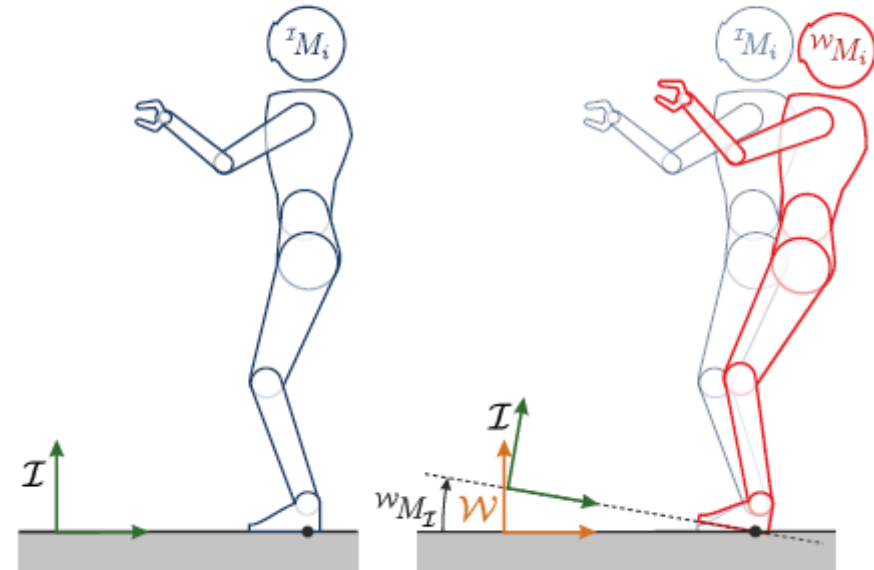
ARTIFICIAL SKIN

State of the Art

MODEL-BASED EXTERNAL FORCE/MOMENT ESTIMATION FOR HUMANOID ROBOTS WITH NO TORQUE MEASUREMENT

BY: MEHDI BENALLEGUE (2018)

External force and moment estimated using an inertial mass unit on the robot

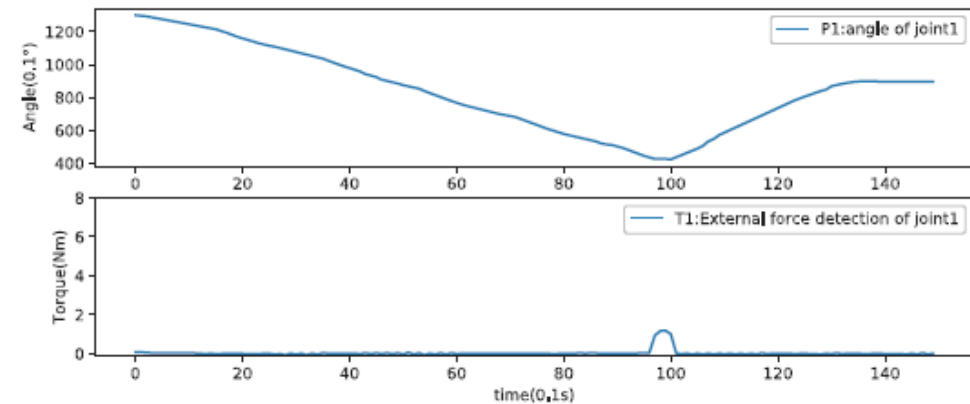


State of the Art

SENSOR-LESS EXTERNAL FORCE DETECTION METHOD FOR HUMANOID ROBOT ARM BASED ON BP NEURAL NETWORK

BY: GUOYU ZUO (2019)

Sensor-less force allocation on a robotic arm using Neural Networks

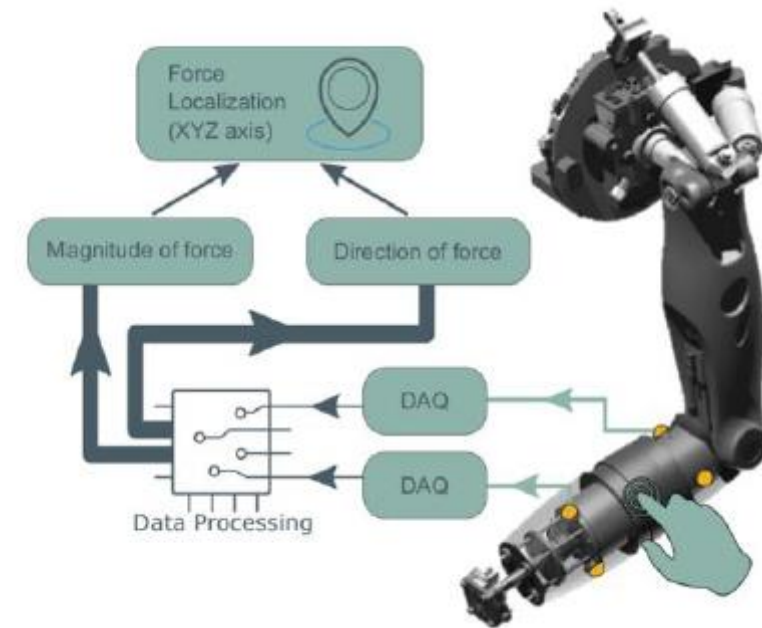


State of the Art

FORCE LOCALIZED INTERACTION SENSING SYSTEM FOR HYDROÏD HUMANOID ROBOT

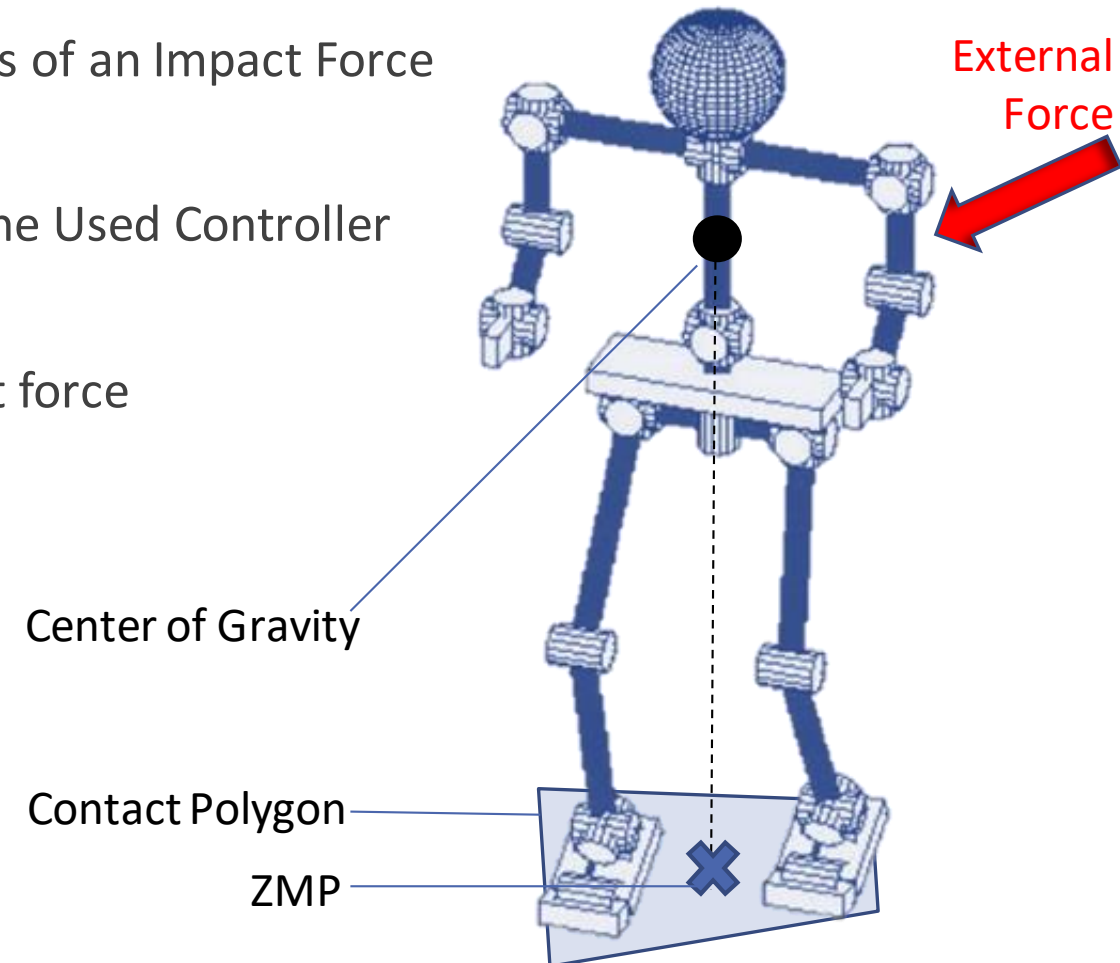
BY: M. TAHOUN (2020)

Force localization based on mechanical deformation and a mathematical model



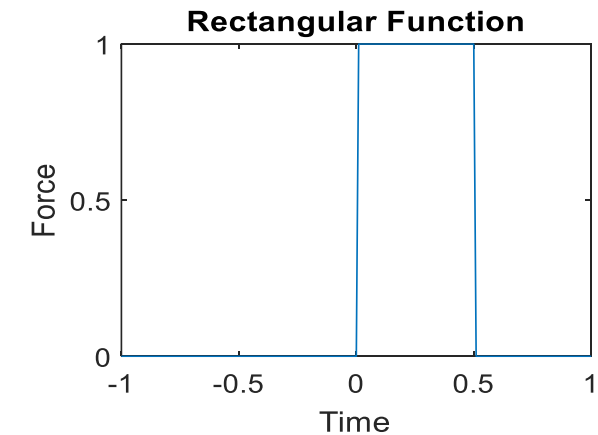
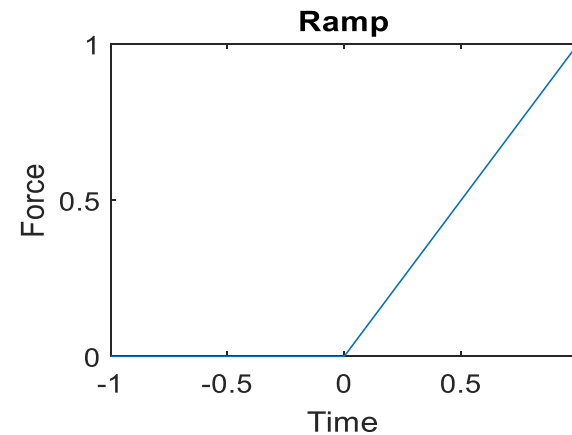
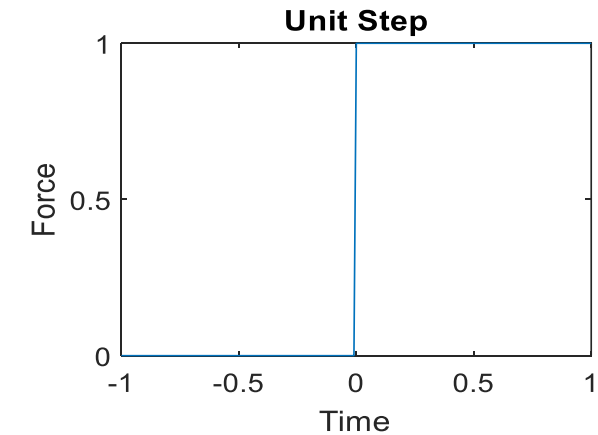
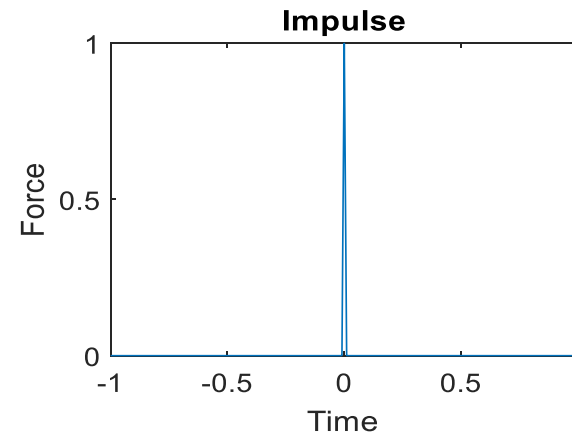
Stability Under Disturbances

- Main Causes of Instability
- Study of the Effects of an Impact Force
- Enhancement of the Used Controller
- Diversity of Impact force



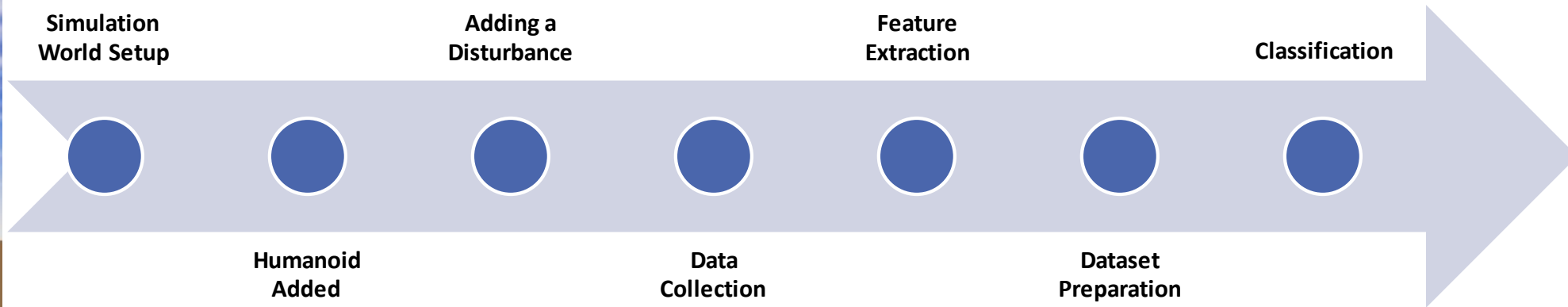
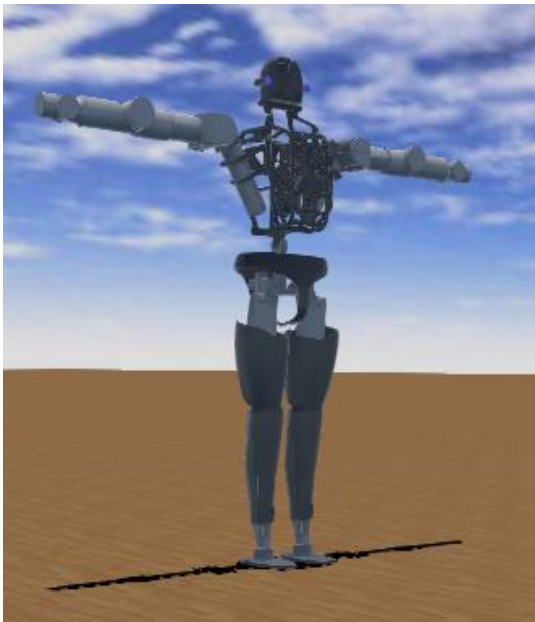
Force Type

- Magnitude
- Orientation
- Location
- Time Duration





Methodology of the Work



Sensors and Solid Nodes Distribution

Neck



- ☐ BackLbz [1]
- ☐ BackMby [2]
- ☐ BackUbx [3]
- ☐ NeckAy [4]

Left Arm



- ☐ LArmElx [5]
- ☐ LArmMwx [6]
- ☐ LArmUwy [7]

Right Arm



- ☐ RArmElx [8]
- ☐ RArmMwx [9]
- ☐ RArmUwy [10]

Right Leg

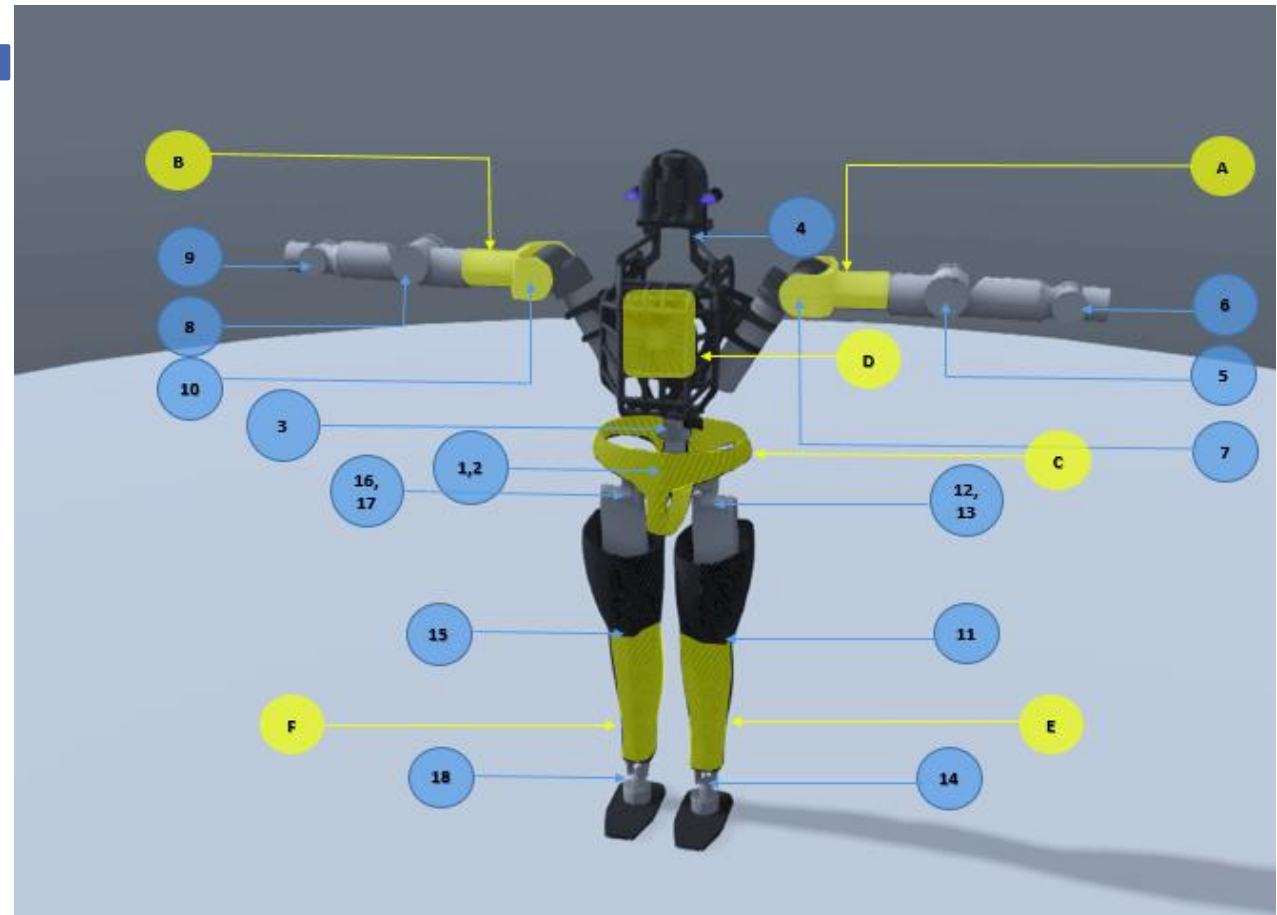


- ☐ RLegKny [15]
- ☐ RLegLhy [16]
- ☐ RLegMhx [17]
- ☐ RLegUay [18]

Solid Nodes



- ☐ Lscap [A]
- ☐ Rscap [B]
- ☐ Pelvis [C]
- ☐ Torso [D]
- ☐ LLL [E]
- ☐ RLL [F]



Impact Force Added

Magnitude

- Ranges from 0 to 2000 N

Orientation

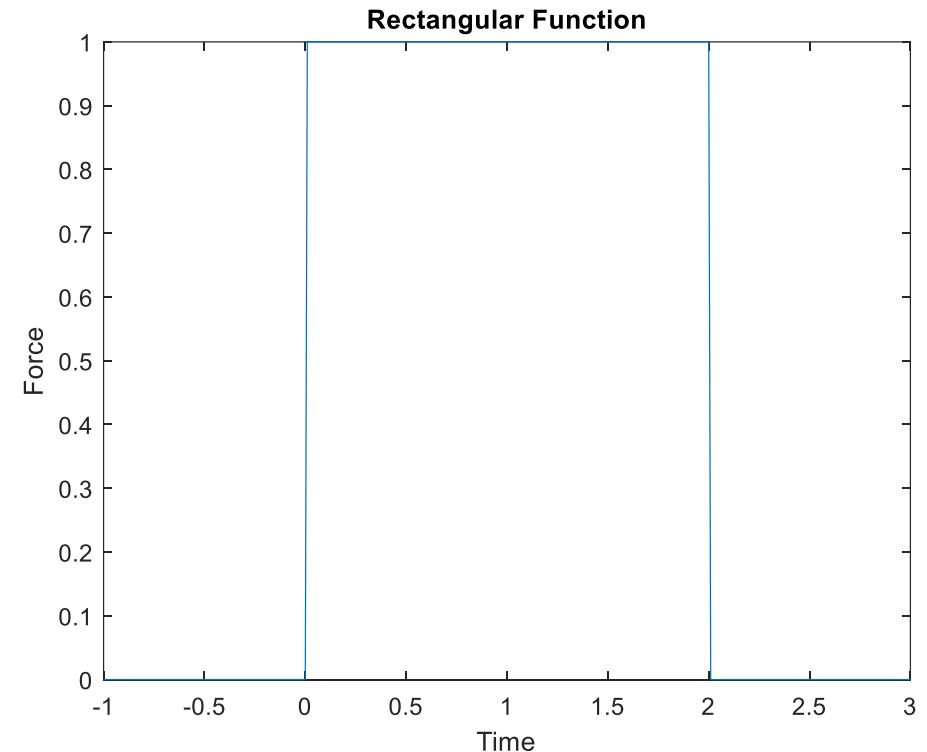
- Along the X-axis (Frontal Direction)

Location

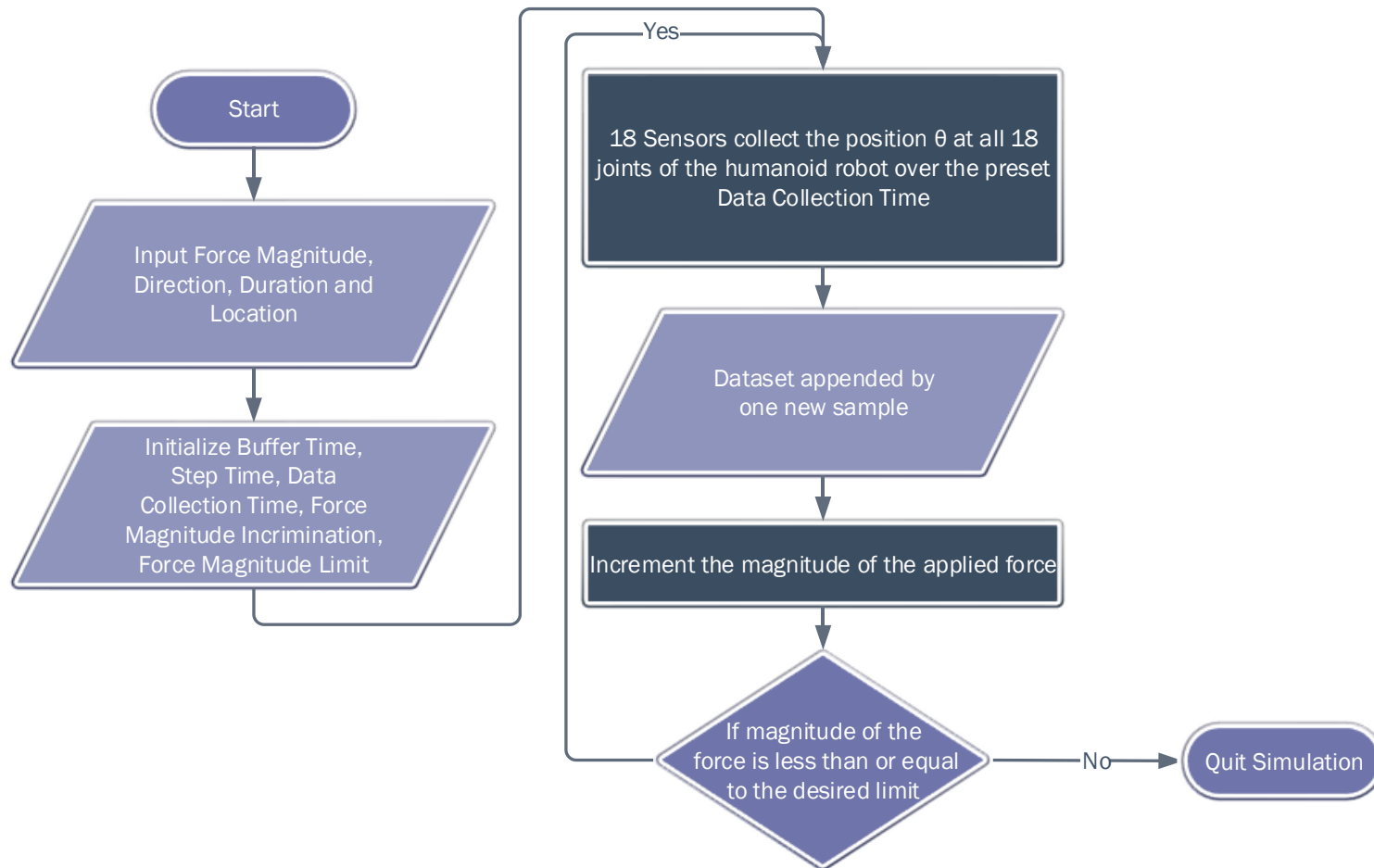
- Center of Gravity of a Solid Node

Time Duration

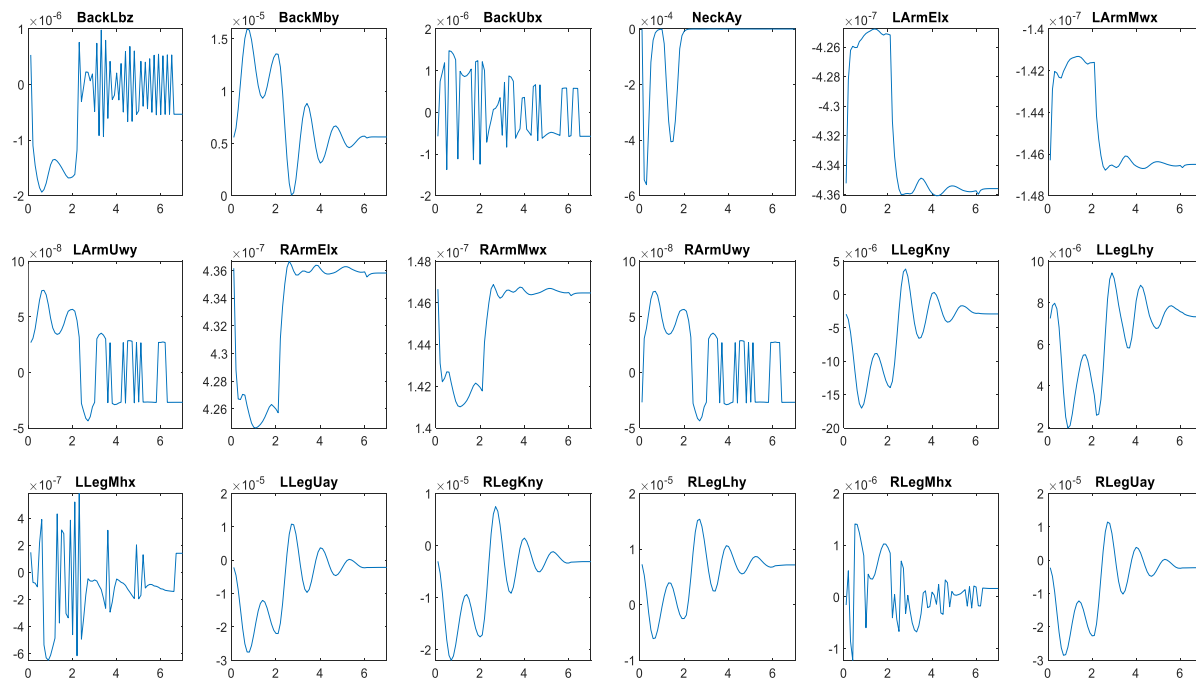
- 2 Seconds



Data Collection



Feature Extraction

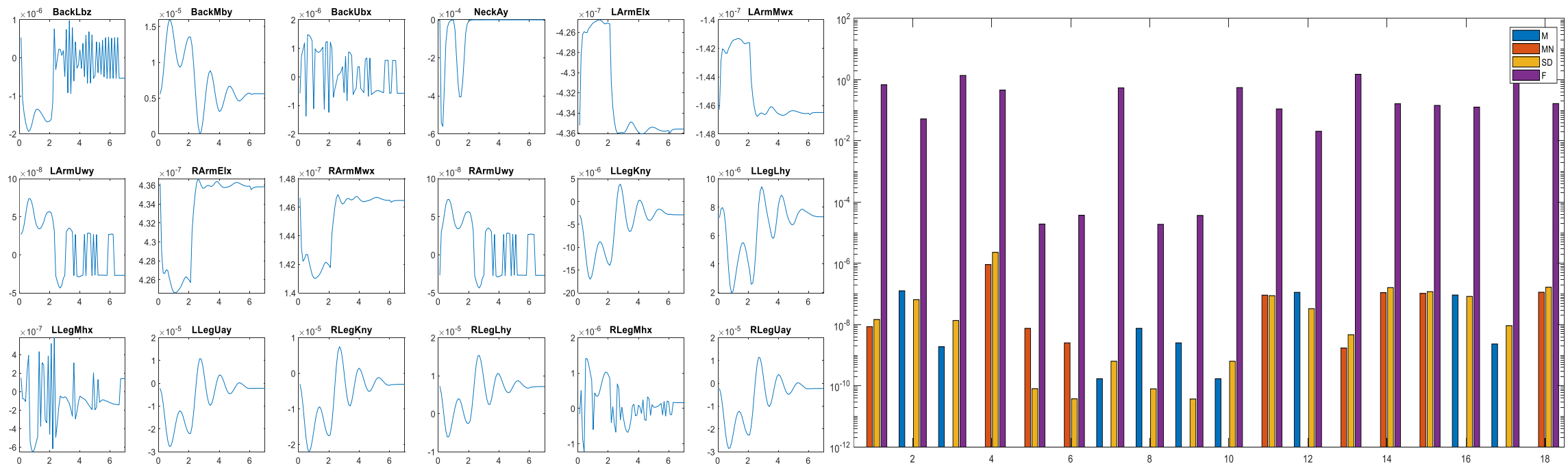


Mean

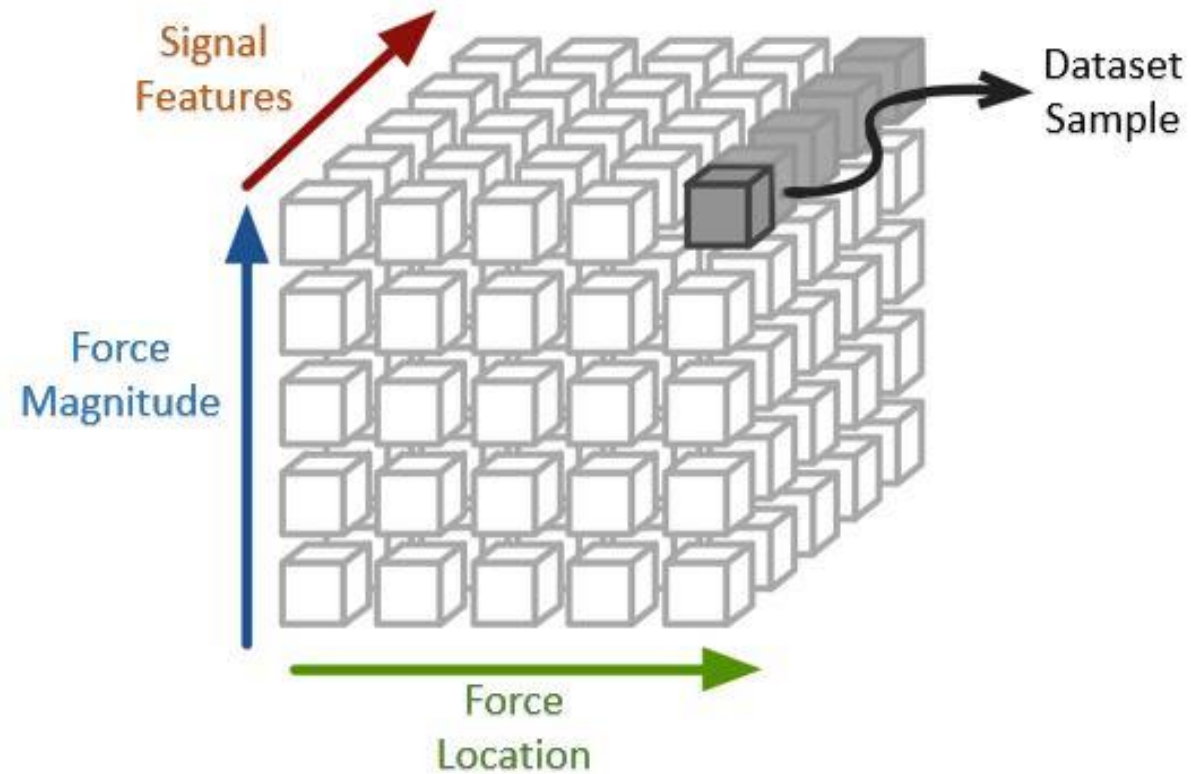
Standard Deviation

Mean Frequency

Feature Extraction



Dataset Representation



Dataset

Samples

- 1300

Features

- 54

Classes for Force Magnitude Range Classification

- 9

Classes for Force Location Classification

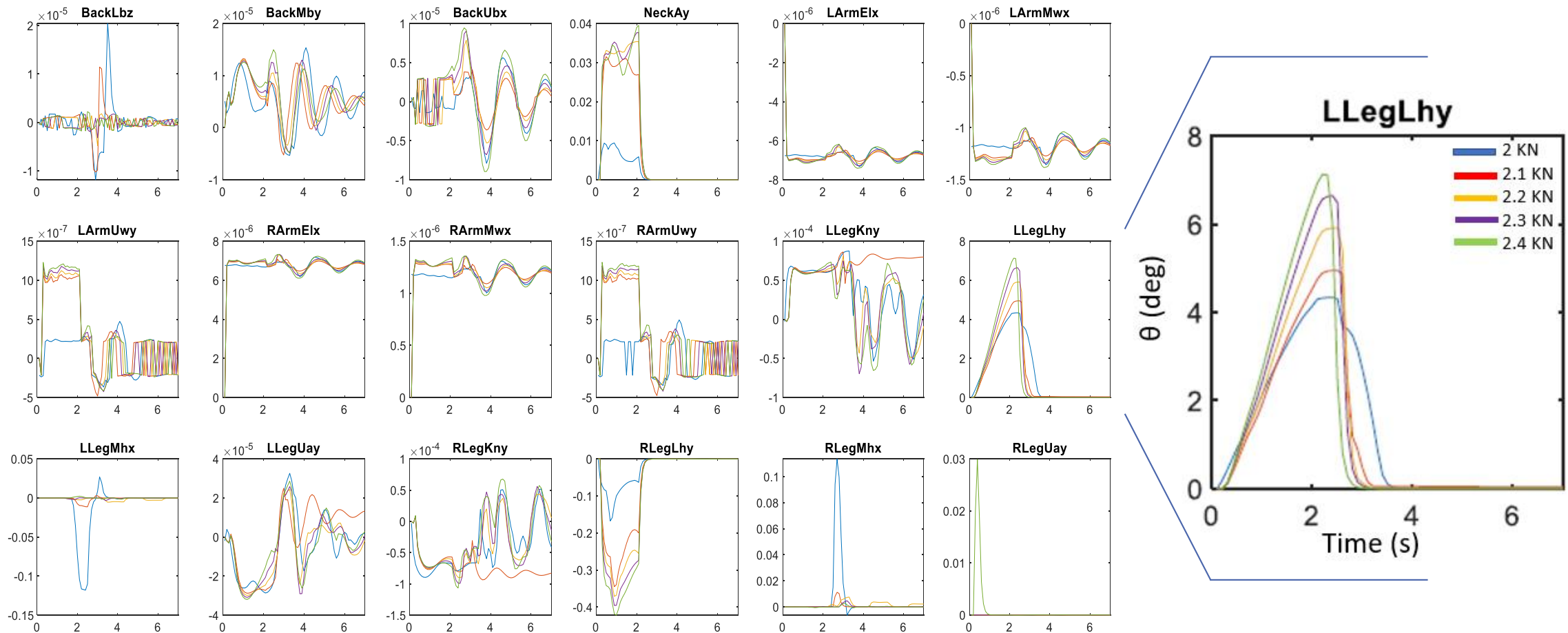
- 6

Classification

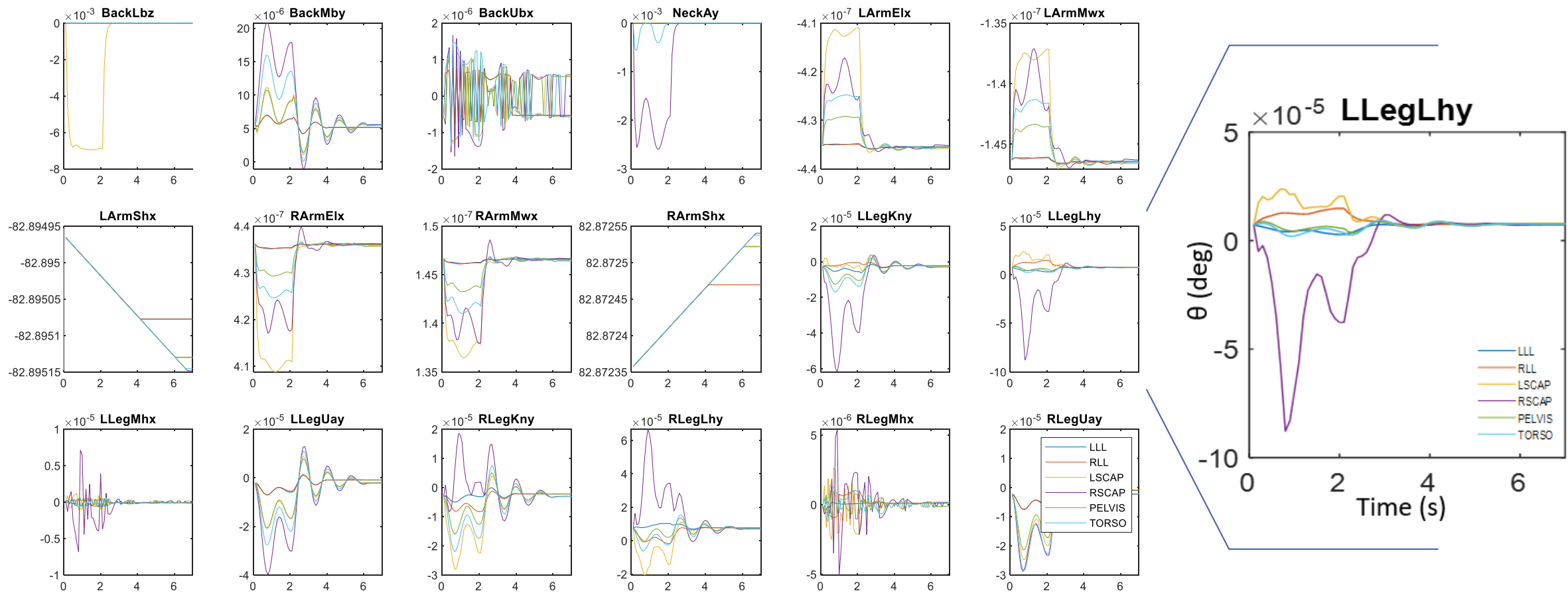
Algorithm	Accuracy (Force Magnitude Classification)	Accuracy (Force Location Classification)
Fine Tree	95.8%	97.2%
Medium Tree	83.8%	96.7%
Coarse Tree	55.2%	73.8%
Linear Discriminant	92.2%	92.6%
Quadratic Discriminant	Failed	94.7%
Linear SVM	90.8%	88.8%
Quadratic SVM	94.6%	95.5%
Cubic SVM	94.2%	96.6%
Fine Gaussian SVM	93.3%	93.2%

Algorithm	Accuracy (Force Magnitude Classification)	Accuracy (Force Location Classification)
Medium Gaussian SVM	91.9%	90.4%
Coarse Gaussian SVM	74.6%	78.2%
Fine KNN	92.8%	93.8%
Medium KNN	89.9%	90.8%
Coarse KNN	65.4%	73.3%
Cubic KNN	90.0%	90.8%
Weighted KNN	87.8%	89.6%
Boosted KNN	92.1%	92.6%
Ensemble Boosted Trees	95.2%	98.2%
Ensemble Bagged Trees	97.3%	98.8%
Ensemble Subspace Discriminant	77.2%	88.1%
Ensemble Subspace KNN	90.3%	92.2%
Ensemble RUSBoosted Trees	91.5%	97.5%

Effect of the Force Intensity



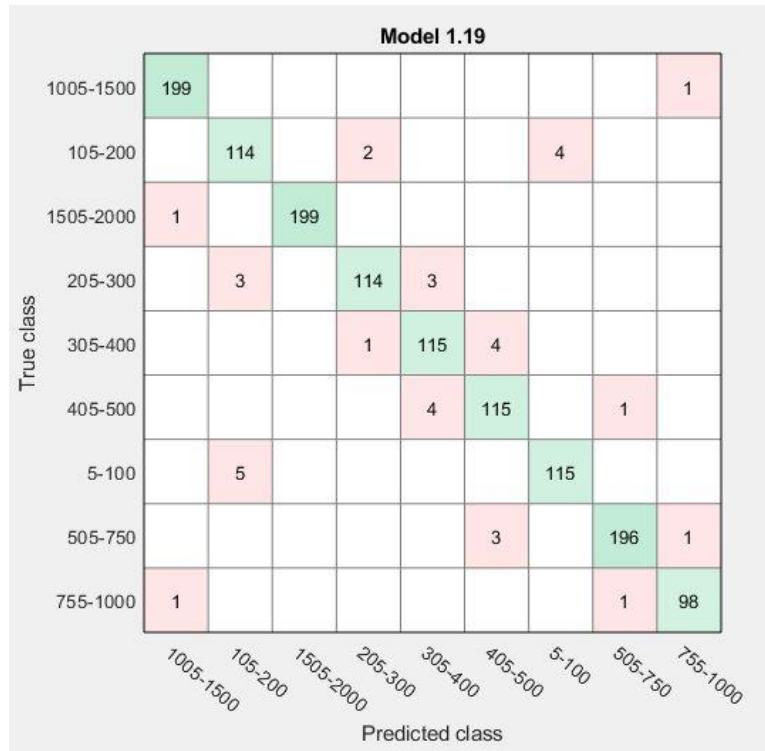
Effect of the Force Location of Application



Ensemble Bagged Trees Algorithm Results

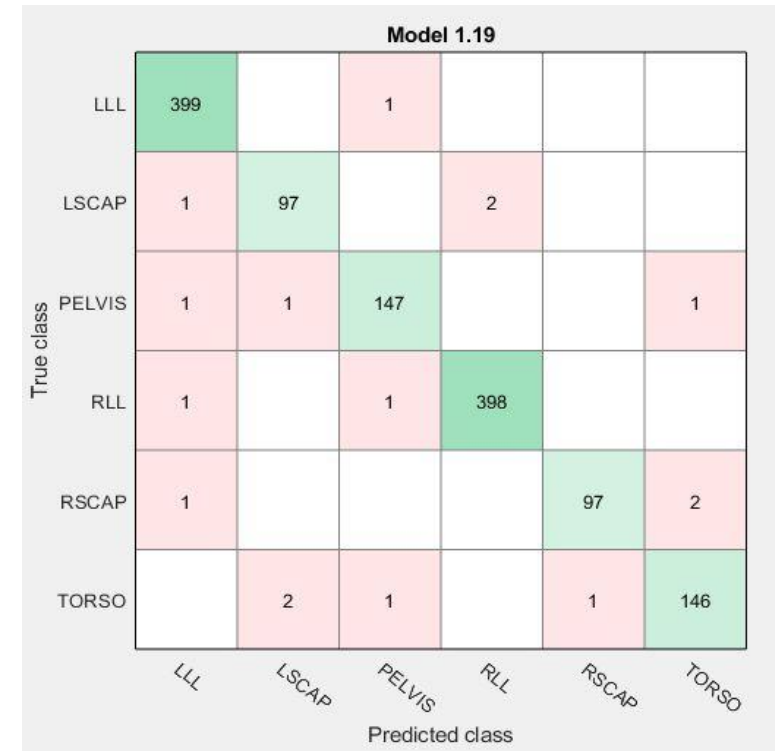
FORCE MAGNITUDE RANGE CLASSIFICATION

97.3 % ACCURACY



FORCE LOCATION CLASSIFICATION

98.8% ACCURACY



Conclusion



Humanoids seem to behave in closer resemblance to humans



Giving the humanoid the ability to evaluate and allocate an impact force enhances the stabilization process



Evaluation and allocating an impact force based on the vibrations in all joints with a high accuracy using the Ensemble Bagged Trees algorithm

Thank You